Depth-Layered Reprojection with Occlusion Inpainting for Automated 2.5D Parallax in LED Volume Production

Olivia Doman

Abstract—LED volume stages have become a dominant tool in modern virtual production, but background plates used on these stages are typically static, requiring full 3D reconstruction pipelines to introduce camera-reactive parallax. We investigate whether a simplified layered 2.5D reprojection pipeline can produce convincing parallax from a single image at production scale, without per-shot manual intervention. Our pipeline automatically estimates depth, partitions the scene into discrete depth layers, synthesizes occluded regions via generative inpainting, and composites the result in a real-time rendering environment.
Our main finding is that automated 2.5D reprojection produces perceptually convincing parallax for LED volume use when camera translation remains below approximately 80 pixels; beyond this threshold, hard depth-threshold boundaries and inpainting seams become clearly visible. We evaluate three monocular depth estimation models—Depth Anything V2, MiDaS, and ZoeDepth—and find that Depth Anything V2 produces the cleanest layer boundaries. The automated pipeline completes depth estimation, segmentation, and inpainting in 2–5 minutes per plate on a consumer GPU, making it viable for production-scale background plate processing under constrained camera movement.

Index Terms—Virtual Production, Depth Estimation, Image Inpainting, LED Volume, Parallax, Layered Depth Images

1 Introduction

LED volume stages—large-format LED displays used as in-camera virtual environments—have transformed virtual production workflows in film and television. By replacing traditional green screen compositing with real-time rendered backgrounds displayed on physical LED walls, productions achieve realistic lighting interaction, reduced compositing costs, and immediate on-set feedback. However, background plates displayed on LED volumes are typically static: they do not respond to camera movement with the parallax that would be expected from a real three-dimensional environment.

Introducing true parallax to an LED volume background currently requires either full 3D reconstruction of the environment or manual artist intervention on a per-shot basis. For productions working with a large number of background plates, neither approach scales well. Manual layer extraction by a skilled compositor can take hours per plate, making sequence-level automation highly desirable.

Because many LED volume shots use limited lateral camera translation, the relevant benchmark is not perfect novel-view synthesis but artifact-free parallax within a narrow motion range. This production constraint is precisely what makes a simplified 2.5D approach defensible: rather than reconstructing true scene geometry, we ask only whether layered reprojection is *good enough* for the camera movement ranges actually used on set.

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We investigate whether a simplified layered 2.5D reprojection pipeline can produce convincing parallax at production scale. Our central finding is that it can—under constraints. Automated depth-based layer extraction and generative inpainting produce results that are perceptually production-plausible for camera translations up to approximately 80 pixels, with Depth Anything V2 producing the strongest layer masks. Hard depth thresholding remains the primary technical bottleneck: even with the strongest depth model, converting continuous depth estimates into binary masks introduces boundary artifacts that grow more visible as camera offset increases.

The central research question is: *can automated depth-based layer extraction and generative inpainting produce parallax results that are perceptually convincing for LED volume production use, and if so, under what constraints?*

This paper makes the following contributions:

- A complete automated pipeline for 2.5D parallax generation from a single image, integrating monocular depth estimation, depth-to-layer segmentation, and generative inpainting.
- A systematic evaluation of three depth estimation models and two inpainting approaches on a representative exterior background plate.
- A characterization of the camera motion threshold beyond which layered reprojection produces objectionable artifacts.

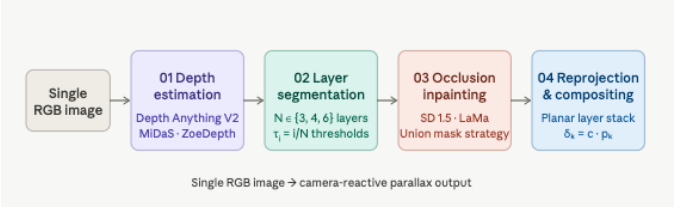


Fig. 1: Overview of the automated 2.5D parallax pipeline. A single RGB image is processed through four stages: monocular depth estimation, depth-to-layer segmentation, per-layer occlusion inpainting, and real-time reprojection and compositing.



Fig. 2: Example input image used in our evaluation: a winter forest exterior scene with clear foreground, midground, and background depth zones.

2 Related Work

2.1 Layered Depth Representations

Shade et al. [1] introduced Layered Depth Images (LDIs), enabling novel view synthesis from a restricted viewpoint range without full 3D reconstruction. Shih et al. [2] presented a complete system for 3D photography from a single image using context-aware inpainting to synthesize occluded regions. Our work differs in focus: we evaluate whether a simplified version of this approach is viable for LED volume production, where camera motion is constrained and per-plate automation is a primary concern.

2.2 Monocular Depth Estimation

MiDaS [3] introduced mixed-dataset training producing relative depth maps that generalize across scene types. Depth Anything V2 [4] builds on this with improved fine detail recovery. ZoeDepth [5] extends relative depth estimation with metric scale prediction. We evaluate all three in our pipeline.

2.3 Image Inpainting

LaMa [6] demonstrates that Fourier convolution-based inpainting achieves robust structural coherence on large mask regions. Stable Diffusion inpainting [7] provides richer textural generation at increased computational cost. We evaluate both approaches.

3 Method

Our pipeline consists of four stages: depth estimation, layer segmentation, occlusion inpainting, and real-time reprojection and compositing, as illustrated in Figure 1.

3.1 Depth Estimation

Given a single RGB input image $I \in \mathbb{R}^{H \times W \times 3}$, we estimate a dense relative depth map $D \in \mathbb{R}^{H \times W}$ using a pre-trained monocular depth estimation model. Depth values are normalized to $[0, 1]$, where higher values indicate greater distance from the camera. Figure 3 shows representative depth maps produced by each of the three models on the same input image. All depth estimation is performed offline within a ComfyUI node workflow.

3.2 Layer Segmentation

We partition the normalized depth map D into N discrete binary masks $\{M_0, M_1, \ldots, M_{N-1}\}$ representing depth-ordered layers from background to foreground. Layer boundaries are defined by evenly-spaced depth thresholds $\tau_i = i/N$ for $i = 1, \ldots, N-1$, such that pixel (x, y) is assigned to layer k if $\tau_{k-1} \leq D(x, y) < \tau_k$. We evaluate layer counts $N \in \{3, 4, 6\}$. Segmentation is implemented as a standalone Python script using OpenCV.

Figure 5 shows the binary masks produced by each depth model across the three layer configuration ($N = 3$). Figure 4 shows the corresponding image regions isolated by each mask when applied to the original image.

3.3 Occlusion Inpainting

For each layer k, we synthesize content in regions occluded by foreground layers using a generative inpainting model. The inpainting mask for layer k is defined as the union of all foreground layer masks $\bigcup_{j>k} M_j$. This cumulative mask strategy is critical: background layers receive the combined midground and foreground mask, midground receives the foreground mask only, and the foreground layer requires no inpainting. Without this union approach, background layers retain painted foreground objects that become visible as ghosting when layers shift during camera movement.

We evaluate LaMa, which provides fast and structurally coherent fills, and Stable Diffusion inpainting (SD 1.5), which produces richer textural variation. Scene-appropriate prompts guide the SD model toward content consistent with the visible scene.

3.4 Reprojection and Compositing

The layered representation is rendered in real time. Each layer is instantiated as a planar mesh at a depth-proportional Z offset from the camera, assigned a translucent material with its corresponding RGBA texture. Camera lateral translation produces apparent parallax through perspective projection: layers at greater Z depth appear to move more slowly relative to the camera, producing the expected depth cue. Parallax offset per layer is given by $\delta_k = c \cdot p_k$, where c is the camera translation and $p_k \in [0, 1]$ is the per-layer parallax factor.

4 Experimental Results

4.1 Setup

We evaluate our pipeline on a single winter forest exterior scene (Figure 2), chosen for its range of depth zones—clear

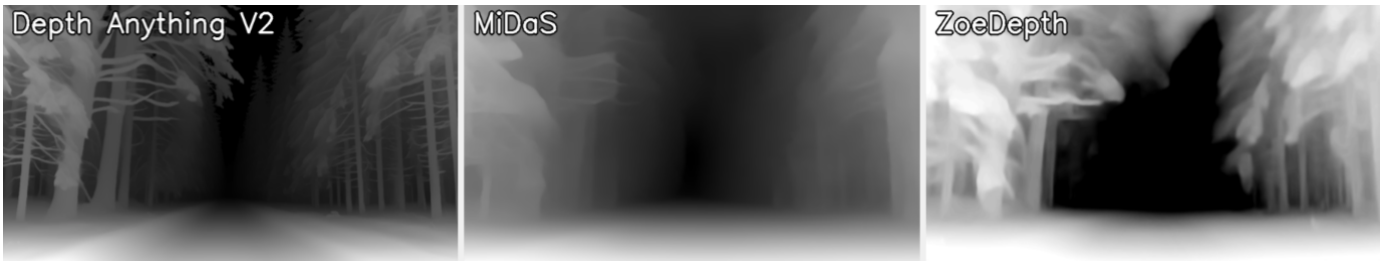


Fig. 3: Depth maps produced by three monocular depth estimation models on the same input image. Depth Anything V2 (left) preserves fine structural detail in the tree canopy. MiDaS (center) produces smoother but less detailed estimates. ZoeDepth (right) shows stronger contrast between near and far regions but with some loss of fine edge detail.

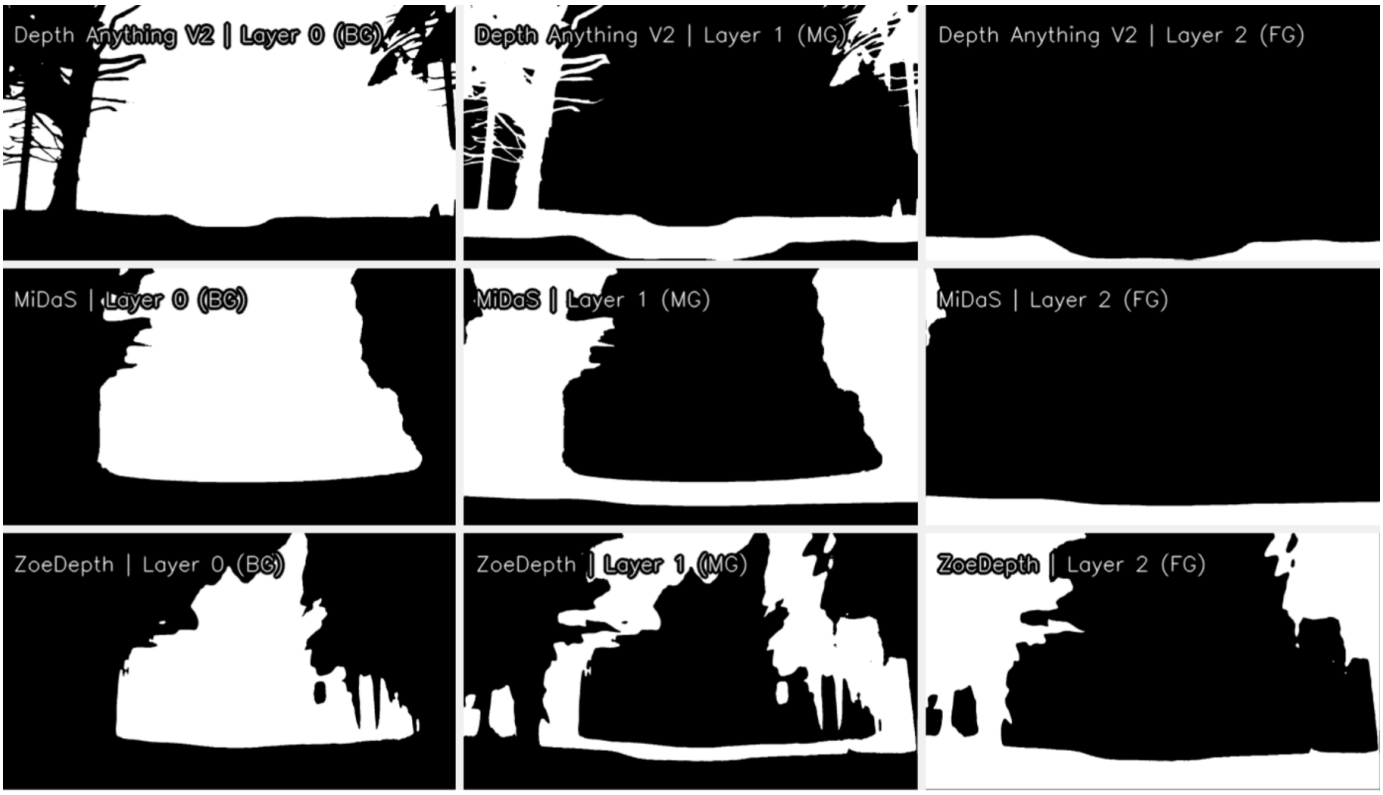


Fig. 4: Binary layer masks ($N = 3$) produced by each depth model on the same input image. Each row shows background (layer 0), midground (layer 1), and foreground (layer 2) masks for one model. Depth Anything V2 (top) produces cleaner mask boundaries with better preservation of thin structures such as tree branches. MiDaS (middle) produces smoother but spatially blurred boundaries. ZoeDepth (bottom) shows stronger depth contrast but noisier segmentation at layer boundaries.

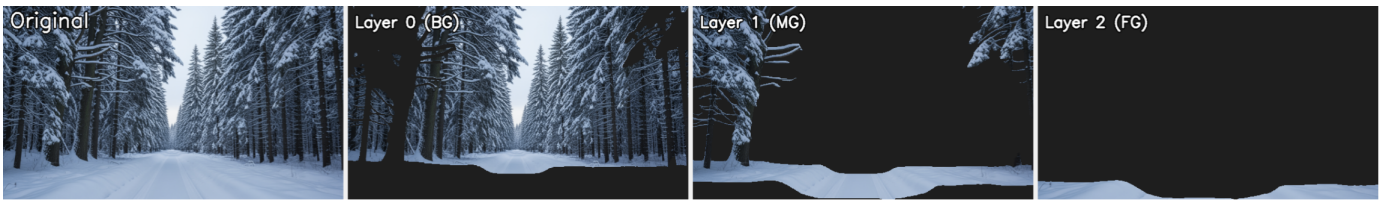


Fig. 5: Layer cutouts produced by applying Depth Anything V2 masks to the original image. From left: original image, background layer (sky and distant trees), midground layer (mid-distance trees and road center), foreground layer (close foreground road and near trees). Dark regions indicate transparent areas to be filled by inpainting.

foreground trees, a midground road, and a distant sky—representative of typical LED volume background plates. Generalization to other scene types (interiors, urban, scenes with human subjects) remains future work.

We evaluate the following variables: depth model (Depth Anything V2, MiDaS, ZoeDepth), layer count ($N \in$

$\{3, 4, 6\}$), inpainting method (LaMa, SD 1.5), and camera translation magnitude (small, medium, large).

We define a result as *production-plausible* when parallax is visible during small camera translations and no artifact immediately draws attention to layer boundaries, missing occluded content, or incorrect depth ordering.

4.2 Depth Model Comparison

Figure 3 shows qualitative depth map outputs from each model. Depth Anything V2 produces the most detailed depth estimates, preserving fine structure in the tree canopy and individual branches. MiDaS produces smoother estimates that blur fine depth discontinuities, resulting in noisier layer boundaries as visible in Figure 5. ZoeDepth shows strong foreground-background contrast but with more fragmented midground segmentation.

Depth Anything V2 is the strongest candidate for automated layer segmentation, producing cleaner mask boundaries and better preservation of fine scene structure compared to MiDaS and ZoeDepth.

4.3 Camera Motion Threshold

We evaluate parallax output at three camera translation magnitudes. At small offsets ($\leq$ 30 px), layer edges are undetectable and results are fully production-ready with no visible artifacts. At medium offsets (30–80 px), slight edge fringing appears at depth boundaries but remains acceptable for wide shots or fast-motion sequences where artifacts are less perceptible. At large offsets ($>$ 80 px), hard cutout edges and inpainting seams become clearly visible, making results unsuitable for production use without further processing.

We therefore define the production-plausible motion range for this pipeline as camera translations up to approximately 80 px. This threshold aligns with the constrained camera movement typical of LED volume interior shots, suggesting the pipeline is viable for a meaningful subset of production use cases.

4.4 Inpainting Comparison

We use Stable Diffusion inpainting (SD 1.5) via ComfyUI as our primary inpainting method, using the cumulative union mask strategy described in Section 3.3. This approach eliminates the ghosting artifacts produced when individual layer masks are used in isolation.

LaMa produces faster fills with stronger structural coherence on large uniform regions such as sky and road, but generates less convincing texture variation in organic areas such as forest undergrowth and mid-distance tree coverage. SD 1.5 produces richer, more scene-consistent fills in these regions at the cost of longer per-layer processing time. For the winter forest scene, SD 1.5 inpainting produced more convincing background reconstruction in the regions behind foreground trees.

The primary inpainting failure mode observed is texture mismatch in filled regions: the inpainted content is locally plausible but inconsistent with the surrounding visible scene in color temperature or surface texture, becoming apparent when layers shift during camera movement.

5 Discussion

5.1 Summary of Findings

The qualitative and quantitative results together support a qualified positive answer to the central research question. Automated 2.5D reprojection can produce production-plausible parallax for LED volume background plates under constrained camera movement, with Depth Anything V2 as the recommended depth model and SD 1.5 inpainting with cumulative union masks as the recommended inpainting strategy.

5.2 Failure Analysis

The following failure modes were observed across the evaluation:

Depth boundary noise. Thin structures such as tree branches and canopy edges produce fragmented mask boundaries in all three depth models. This is the most common failure mode and is visible at even small camera translations when the structure is in the foreground layer.

Inpainting texture mismatch. Filled regions behind foreground layers are locally plausible but sometimes inconsistent with the surrounding scene in color or texture. This becomes apparent when layers shift during playback.

Hard boundary halos. At medium and large camera translations, hard depth-threshold boundaries produce visible edge halos around layer cutouts. This is inherent to binary mask segmentation and is not fully resolved by any of the three depth models tested.

Incorrect midground separation. Evenly-spaced depth thresholds occasionally assign perceptually coherent objects—such as a continuous road surface—to different layers, producing unnatural splits visible during camera movement.

Background ghosting (prior approach). An earlier version of the pipeline using per-layer masks (rather than cumulative union masks) produced ghosting artifacts where foreground objects remained visible on the background layer during large camera offsets. This was resolved by the union mask strategy described in Section 3.3.

5.3 Limitations

The evaluation is limited to a single scene type (winter forest exterior). Generalization to scenes with more complex depth distributions, human subjects, or curved LED volume configurations has not been validated. The 80 px motion threshold is specific to this scene and pipeline configuration; scenes with thinner foreground structures or more complex midground content may produce objectionable artifacts at smaller offsets.

6 Conclusion

We presented an automated pipeline for generating 2.5D parallax from single images using depth-layered reprojection and generative inpainting, motivated by the need for scalable parallax-enabled background plates in LED volume production.

Our central finding is that the pipeline produces perceptually convincing parallax for camera translations up

TABLE 1: Depth Model Comparison

Model	Edge Quality	Layer Sep.	Runtime (s)
Depth Anything V2	High, fine branches	Clean boundaries	3.2
MiDaS	Moderate, blurred	Noisy at edges	2.1
ZoeDepth	High contrast, coarse	Fragmented midground	4.8



(a) Small (25 px). No visible artifacts.

(b) Medium (75,px). Slight edge fringing.

(c) Large (150 px). Hard cutout edges visible.

Fig. 6: Parallax output at increasing camera translation magnitudes using Depth Anything V2 with $N = 3$ layers. Artifacts become objectionable beyond approximately 80 px of lateral translation.



(a) DAV2 — Background

(b) DAV2 — Midground

(c) DAV2 — Foreground

(d) MiDaS — Background

(e) MiDaS — Midground

(f) MiDaS — Foreground

(g) ZoeDepth — Background

(h) ZoeDepth — Midground

(i) ZoeDepth — Foreground

Fig. 7: Inpainting results by depth model and layer. Each image shows the scene with that layer's content removed and the occluded region filled by SD 1.5 inpainting. Background layers (column 1) use the combined midground and foreground mask; midground layers (column 2) use the foreground mask only. Depth Anything V2 (top row) produces the most coherent fills owing to cleaner input masks.

to approximately 80 px, with no visible artifacts at small offsets ($\leq$ 30 px) and acceptable results at medium offsets (30–80 px). Depth Anything V2 produces the strongest layer masks, and cumulative union mask inpainting eliminates the ghosting artifacts present in simpler per-layer approaches. The automated pipeline completes depth estimation, segmentation, and inpainting in 2–5 minutes per plate on a consumer GPU, representing a meaningful reduction

in per-plate processing time relative to manual compositor workflows.

Hard depth-threshold segmentation remains the primary limitation: converting continuous depth estimates into binary masks introduces boundary artifacts that grow more visible with camera translation. Future work should investigate soft-edge layer blending with Gaussian-feathered alpha masks to reduce edge halos, scene-adaptive depth thresholds to better match natural scene structure, and extension to vertical camera movement and curved LED volume configurations.

Acknowledgments

The author thanks Kayvon Fatahalian for sponsoring this project.

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